

Term Project - Milestone 4

DSC640-T301

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I focused on understanding economic conditions and childcare pricing trends across states and the years within the dataset. The key variables are poverty rates for families (PR_F), median household income (MHI), and median childcare prices across infant, toddler, and preschool categories (MFCCInfant, MFCCToddler, MFCCPreschool). Analysis revealed several patterns and relationships. One is that the childcare costs vary widely by state, with some states doubling the cost of others. This suggests that geographic economic conditions or policies significantly impact childcare affordability. Median household income correlates positively with childcare costs across all age groups, suggesting that cost-of-living patterns exist in areas where childcare providers operate in higher-wage markets. All childcare price categories showed strong positive correlations, indicating that childcare prices are impacted across age groups. There is also a demonstrated gradual increase in childcare costs over time, with potential outliers or increased variance. Some assumptions that were made during this analysis included that the reported childcare prices represent statewide medians. We also assume that cost-of-living or inflation adjustment is not embedded in the dataset. Further clarification on several items will still be necessary for a more comprehensive analysis, including consistent datasets across all analyzed states. The technical guide already stated that assumptions were made and values were entered for specific fields related to childcare prices across age groups and statewide relationships.

The plan of attack is to present the story of rising childcare costs and the burden they are placing on families across the United States, and use this to drive for government intervention and childcare reform. The target audience would be the public, aiming to drive this as a key factor in elections and regulation conversations. As such, the data and presentation need to be kept at a high level but direct and impactful. The public's attention span is short, and it needs to be captured. As a result, I plan to include an infographic, a dashboard, and short-form videos (TikTok, Instagram). These are media that can be placed in front of the public and communicate ideas efficiently to those who do not intend to digest new information. They are commonly encountered and should facilitate communication with the audience. To follow this intent, the visualizations chosen for each medium were designed to reinforce quick, easy digestion. The short-form video featured a statistical callout followed by a line graph in motion, allowing for quick trend understanding without spending too much time digesting the content. The infographic centered on a map of the U.S., again something recognizable to the audience and likely to drive engagement

by encouraging them to connect their personal stories to the data based on location. This is then supplemented with additional statistics drawn from the data to reinforce the impact of childcare costs. The dashboard was then built to build on that story investment, based on location, by allowing filtering by state to drive to their local childcare information. Across all media, the color scheme remained consistent, with childcare costs shown in shades of red and total income in blues. Statistical callouts and values were also kept consistent with the state's median household income across all mediums to ensure consistent messaging and understanding.

This analysis required several data preparation and transformation steps that carry ethical implications related to accuracy, transparency, and public interpretation. State-level medians were calculated for household income, poverty rates, and childcare costs to enable consistent comparisons across geographic regions. Weekly childcare price measures were annualized to align with annual household income again to enable consistent comparisons and reduce audience mental load. Comparisons of percentage cost change were based on the earliest and latest years with valid observations for each state to avoid distortions from missing data. While these transformations improve interpretability, they also further aggregate already summarized public data, which may obscure within-state variation and limit representation of specific populations.

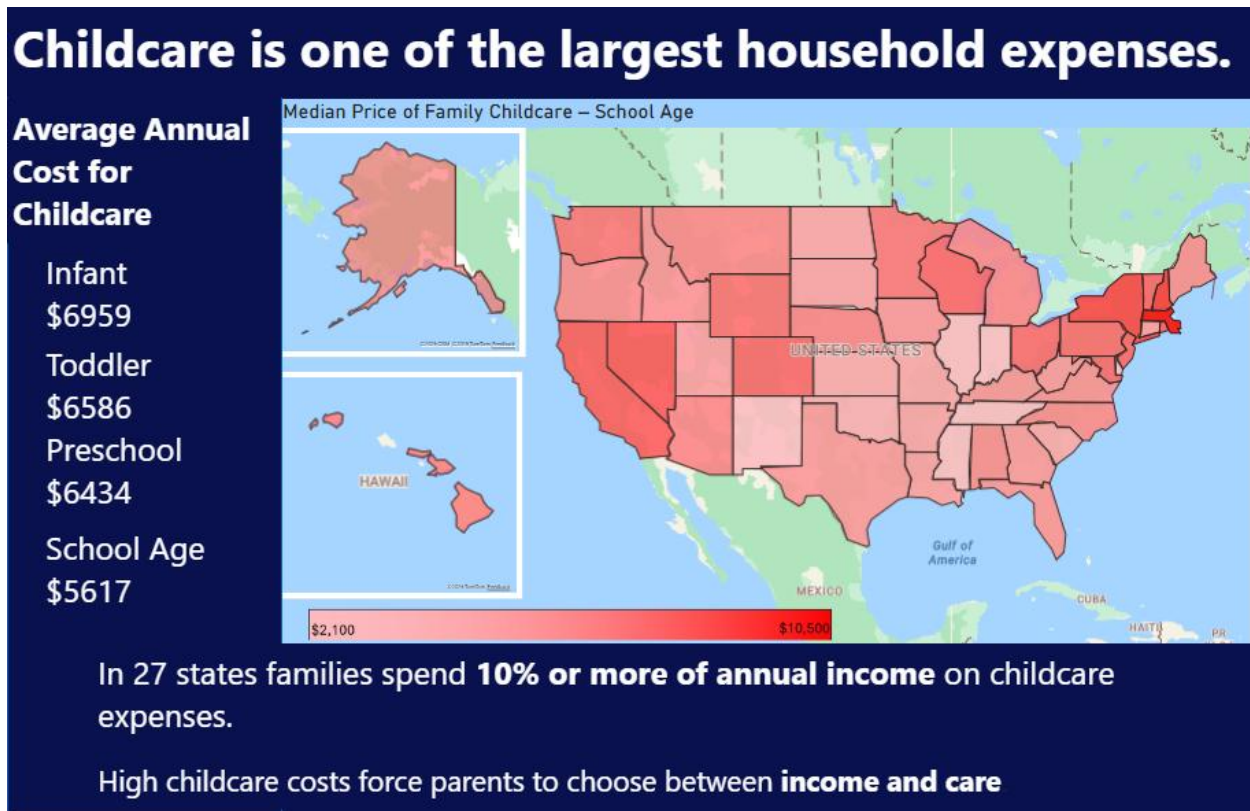
The dataset originates from publicly available governmental and research sources and contains only aggregated, non-identifiable information. As a result, the analysis does not implicate privacy regulations such as HIPAA or FERPA, and the data can be considered ethically acquired and appropriate for public policy discussion. However, ethical responsibility remains in how findings are framed and communicated, particularly because the intended audience is the general public and the visualizations are designed to influence civic awareness and policy discourse.

Additional risks arise from how results are framed and visualized for a general public audience. The project intentionally aims to inform public opinion and policy conversations and there is a risk that simplified visuals or statistical callouts could be interpreted as causal claims or could overstate the burden across all families. The use of maps, color gradients, and short-form video also increases the potential for emotional framing, which may influence interpretation beyond what the data alone supports.

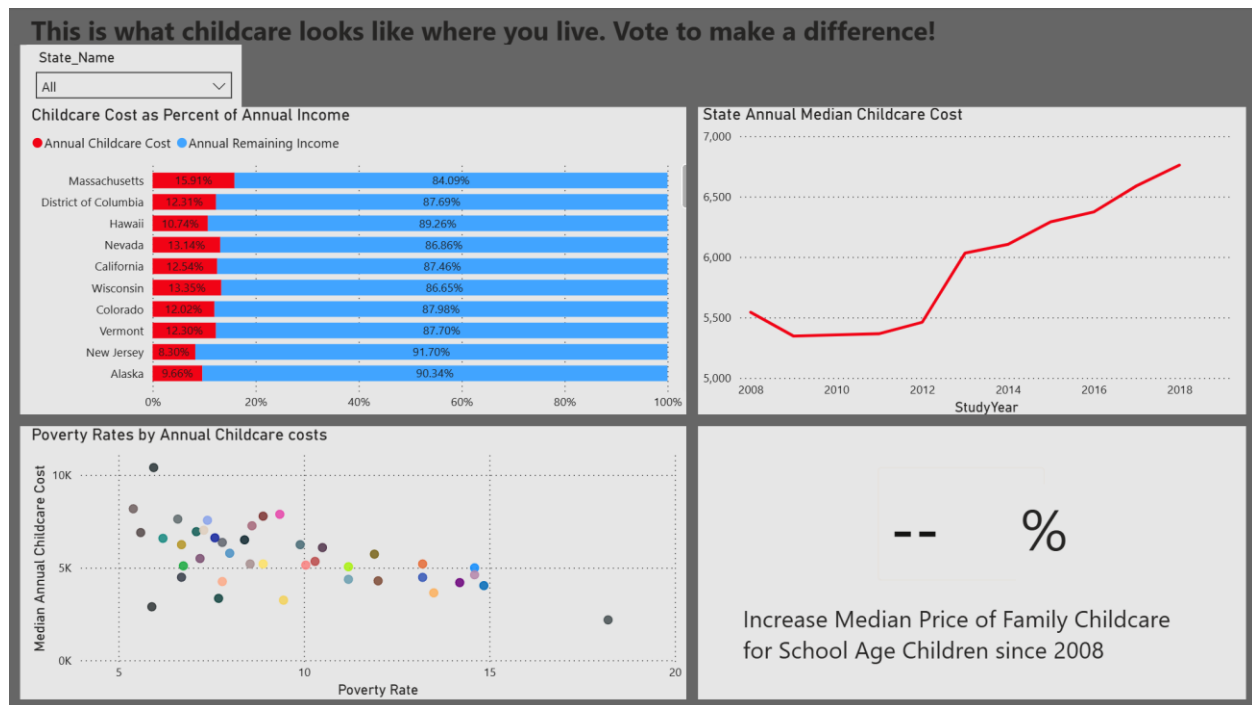
Potential risks include oversimplification of complex economic relationships, unintended implication of causal conclusions from correlational patterns, and visual framing choices—such as color intensity, statistic selection, or geographic emphasis—that may heighten emotional response beyond what the data alone supports. To mitigate these risks, all assumptions, aggregation methods, and data limitations are explicitly

documented; medians are used instead of extreme values; causal claims are avoided; and viewers are directed to an interactive dashboard that provides a fuller context rather than relying solely on isolated statistics. These steps aim to balance accessibility with analytical integrity, ensuring that efforts to communicate the burden of childcare costs remain informative rather than misleading.

Infographic



Dashboard (No selection)



Dashboard (State Selected)

